FACULTY OF SCIENCES OF SFAX — UNIVERSITY OF SFAX

A First QLoRA Fine-Tuning Experiment for Infrastructure-as-Code Security Remediation

A preliminary experiment on Gemma-2-2B-IT

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Gemma-2 · QLoRA · IaC Security · Kaggle

Outline

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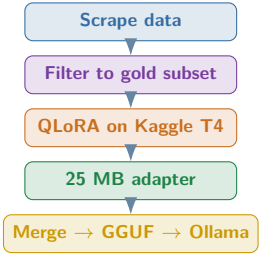


How the research line evolved

What prior work gives us

- A vocabulary of IaC security smells.
- Static and semantic detectors.
- RAG/self-verification patterns.

Where this thesis is positioned

Move from **detecting** smells to generating and validating concrete IaC remediations inside an agentic loop.

Where this work fits in the project

Agentic IaC security framework (Rapport 2)

Seven modules cooperating under a central orchestrator:

- Contextual Analyzer
- Knowledge Base + RAG Retriever
- **Fix Generator** → this work
- External Validator (Checkov + KICS)
- Patch Formatter & Explanation

Two informal observations on the baseline

- Off-the-shelf small LLMs (Gemma-3-4B, Gemma-4-e2B) produce plausible fixes but do not always target the declared CWE.
- On CPU, Gemma-4-e2B $\approx$ **320 s / fix** $\Rightarrow$ full eval with self-consistency is impractical on current hardware.

Why touch the Fix Generator now?

It is the *only* module whose quality depends on the underlying LLM rather than on deterministic software.

This report

A first, preliminary attempt to fine-tune a small open-weight model on our own scraped data.

Scope of this experiment

What this is

A **first, exploratory** run to validate the pipeline end-to-end.

In scope

- Small base model (≈ 2 B params)
- Subset of the scraped data (3,227 train)
- 1 epoch, 1 random seed
- 1 prompt template
- Qualitative check on $\mathbf{n = 3}$ samples
- Kaggle free T4 GPU

Out of scope

- No hyperparameter search
- No comparison against baselines
- No downstream task-level metric (scanner pass-rate)
- No multi-seed variance estimate
- No claim of competitive performance

Goal: demonstrate that the pipeline works and surface the obstacles that block a scaled experiment.

LoRA in one slide

Core idea

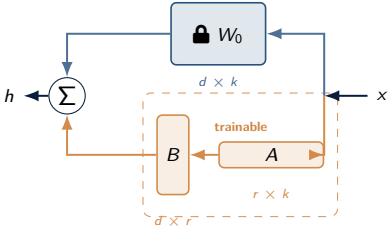
Freeze the pretrained weights $W_0 \in \mathbb{R}^{d\times k}$ and learn a low-rank update

$$\Delta W = BA, \quad B \in \mathbb{R}^{d\times r},\ A \in \mathbb{R}^{r\times k},\ r \ll \min(d,k).$$

Effective weight at inference: $W = W_0 + \Delta W$.

For Gemma-2-2B, $r = 16$ on attention only

- Trainable params: **9.27 M** out of 2.6 B ($\approx 0.35\,\%$)
- Adapter file: **25 MB** (vs. 5 GB full model)



$$h = W_0 x + \frac{\alpha}{r} BAx$$
$$r \ll d, k$$

QLoRA and supervised fine-tuning

QLoRA = LoRA on 4-bit base

- Base weights stored in **NF4** (4-bit NormalFloat) with double-quantised scales
- Tiles decompressed on the fly for forward pass
- Activations in fp16, gradients routed *only* through A/B
- Memory footprint for 2-B model: ≈ 5 GB fp16 $\rightarrow \approx$ **1.6 GB NF4**

Consequence

A 2-B model becomes trainable on a **single 16 GB T4** — exactly what Kaggle's free tier offers.

Supervised fine-tuning (SFT) for code repair

Task: given a *detected smell* and an *insecure snippet*, output a *fixed snippet*.

- Standard causal-LM cross-entropy loss
- Loss computed **only on the model turn** (the fix tokens), prompt tokens are masked
- Closer to code-to-code translation than to free code generation

Why SFT (and not RL / preference optimisation) here?

We have before/after pairs, not ranked preferences. SFT matches the data we already have.

Dataset origin: our own scraper

Five complementary sources

- GitHub **commit search** with date-window partitioning (to bypass the 1,000-result cap)
- **GHArchive** hourly event dumps (bypasses the Search API quota)
- **Scanner test suites** (Checkov / KICS / tfsec): paired "failed" / "passed" examples
- **Known vulnerable repos**: TerraGoat, Kubernetes-Goat, WrongSecrets
- **Scanner-validated labels**: Checkov + tfsec + KICS rerun on every record

Dataset v1 — headline figures

Total records (dedup)	31,748
With a fix	16,362
Scanner-validated	10,791
Source repositories	$>2{,}000$
Smell types	18

Tool mix (v1 raw)

Terraform 50 % ▪ Kubernetes 28 % ▪ Ansible 13 % ▪ Docker 7 % ▪ CloudFormation 1.5 %

From 31,748 raw records to a 3,227-record training set

Filter	Remaining	Share of input
All validated records	31,748	100.0 %
`has_fix` = True	16,362	51.5 %
Scanner-validated only	∼10,791	34.0 %
Non-empty `code_before` and `code_after`	∼10,500	—
Combined length < 8,000 chars	∼5,500	—
Non-noop fix (`before` ≠ `after`)	**4,050**	12.76 %
Pre-existing train split (80 %)	**3,227**	—

Gold subset — better-balanced tool mix

Docker 32 % ▪ Terraform 30 % ▪ Kubernetes 30 % ▪ CloudFormation 5 % ▪ Ansible 3 %

Caveat — this is a deliberate subset

We chose the **strictest** filter for this first run. A larger (but noisier) corpus of up to ∼ 10,791 records is available and has not been used yet.

Training sample format

Slim JSON record

```
{
  "id": "a1b2c3d4",
  "iac_tool": "terraform",
  "smell_types": ["hardcoded_credential", "overly_permissive_cidr"],
  "cwes": ["CWE-798", "CWE-732"],
  "code_before": "resource aws_s3_bucket ...",
  "code_after":  "resource aws_s3_bucket ...",
  "split": "train"
}
```

Rendered at training time via the Gemma-2 chat template

```
You are a security expert specializing in Infrastructure as Code (IaC).
Fix the following {iac_tool} script. Detected smells: {smells}.
Return ONLY the corrected script, no explanation.

‘‘‘
{code_before}
‘‘‘
```

Base model: Gemma-2-2B-IT

Model	Size	Gated	Remark
google/codegemma-2b-it	2 B	Yes	Code-specialised
google/gemma-2-2b-it	2 B	Yes	**Chosen** — matches Ollama stack
Qwen/Qwen2.5-Coder-3B-Instruct	3 B	No	Stronger code model, non-gated
bigcode/starcoder2-3b	3 B	No	No chat template

Reasons for this choice

- Same family as our local Ollama stack (Gemma-3-4B, Gemma-4-e2B) — no new architecture
- Published chat template that maps onto our instruction/response pair
- Smallest Gemma-2: VRAM headroom on T4, a few hours per run

This is a pragmatic choice, not an optimum

No systematic model comparison is in scope here. Larger candidates (3 B, 7 B, 9 B) are deferred to a later iteration.

QLoRA configuration

Parameter	Value
Quantisation	4-bit NF4 (double-quant, fp16 compute)
LoRA rank r	16
LoRA α	32
LoRA dropout	0.05
Target modules	`q_proj`, `k_proj`, `v_proj`, `o_proj`
Trainable params	9,273,344 (0.35 %)
Learning rate	2×10^{-4}
Scheduler / warmup	Cosine / 3 %
Effective batch	8 (1×8 grad. accumulation)
Epochs	**1**
Optimiser	Paged AdamW 8-bit
Max seq. length	2,048 tokens
Precision	fp16 (bf16 unavailable on T4)
Gradient checkpointing	Enabled (non-reentrant)
Random seed	42

Values copied from published recipes — no hyperparameter search was performed in this first run.

Kaggle environment

Runtime

- Accelerator: NVIDIA **T4 (16 GB)**
- CUDA: 12.8
- Python: 3.12
- Session limit: 9 h per notebook
- Internet: on (to fetch base model)
- Attachments: 2 custom Kaggle datasets (scripts + filtered corpus)

Three issues encountered (for reproducibility)

- **403 Forbidden** on Gemma-2
  → accept license on HF page
- `bitsandbytes` 0.43.3 lacks CUDA 12.8 binary
  → upgrade to 0.49.2 in-place
- Kernel restart wipes env vars
  → re-run the HF-token cell first

The five Kaggle cells (summary)

Cell 1 — Install deps

```
!pip install -q -U transformers==4.44.2 \
  datasets==2.21.0 peft==0.12.0 \
  accelerate==0.34.2 bitsandbytes==0.43.3 \
  trl==0.10.1
!pip install -q -U bitsandbytes
```

Cell 2 — Expose HF token

```
from kaggle_secrets import UserSecretsClient
os.environ["HF_TOKEN"] = \
  UserSecretsClient().get_secret("HF_TOKEN")
```

Cell 3 — Smoke test (200 samples)

```
!python 02_train_qlora.py \
  --model google/gemma-2-2b-it \
  --data-dir .../data \
  --output-dir /kaggle/working/lora_smoke \
  --epochs 1 --max-train-samples 200
```

Cell 4 — Full training (1 epoch)

```
!python 02_train_qlora.py \
  --model google/gemma-2-2b-it \
  --data-dir .../data \
  --output-dir /kaggle/working/lora_out \
  --epochs 1 --batch-size 1 \
  --grad-accum 8 --lr 2e-4
```

Cell 5 — Inference check

```
!python 03_inference.py \
  --model google/gemma-2-2b-it \
  --adapter /kaggle/working/lora_out \
  --input-file .../test.jsonl \
  --n-samples 3
```

Smoke test is a sanity check, not a curve

Produces a single final loss, not a learning curve; only the full run yields a per-step loss trajectory.

Training metrics (single run, seed 42)

Metric	Value
Optimisation steps	403
Training wall-clock	1 h 47 min
Evaluation wall-clock	218 s
Average step time	15.94 s
Training throughput	0.502 samples/s
Final training loss	**0.528**
Final evaluation loss	**0.467**
Adapter size (safetensors)	25.6 MB

How to read these numbers

- **One run, one seed** — no confidence interval.
- Absolute loss is tokeniser-dependent; not directly comparable across models or papers.
- `eval_loss` < `train_loss` is consistent with *no overfitting* after 1 epoch, but can also be explained by:
  - training loss averaged over high-loss early steps,
  - LoRA dropout on at train, off at eval,
  - eval split possibly slightly easier.
- **Not** evidence of superior generalisation.

Qualitative inspection — 3 held-out test samples

#	Declared smells	Human ground truth	Fine-tuned model output
1	`apt_no_install_recommends`, `unpinned_base_image` (Docker)	Version bump ATMOS 1.210 → 1.214 — *does not address the declared smell*	Adds `-allow-downgrades`, creates non-root user `appuser` — sensible but unrelated hardening
2	`uri_credentials` (Terraform)	Changes `recovery_window_in_days` 0 → 7 — unrelated	Adds IRSA / IAM scaffolding, keeps 0 — does not target URI credentials
3	`unpinned_base_image` (Docker)	Pins a Go image by `sha256:` digest (✓ real pin)	Does not pin; performs unrelated hardening (`apt-get upgrade`, labels)

What we see (with $n = 3$ caveat)

- Output is syntactically well-formed on these 3 cases.
- Contains idiomatic hardening patterns.

What we see (worrying)

- Model *and human ground truth* often fail to address the declared smell.
- **The label itself is noisy.**

Label noise is the binding constraint

The problem

Commit-scraped pairs are at **file granularity**, but a single commit may bundle several unrelated modifications.

- Regex classifier infers the smell from the *commit message* (not the diff).
- Scanner validation only confirms that `code_before` was vulnerable — not that the *removed lines* address that vulnerability.

Practical consequence

Training signal = average of (a) genuine smell→patch pairs and (b) unrelated refactors that happen to coexist in a "fix: security" commit.
The model can only learn the distribution of *"things that look like fix commits"*.

Two remediations (for next iteration)

1. **Hunk-level filter**: keep only records whose *diff* regex-matches the declared smell.
2. **Scanner-fixture-only corpus**: $\sim$1,850 Checkov + KICS test pairs where the smell is, by construction, what the scanner flags.

Limitations and threats to validity

Experimental limitations

- **Small model** (2 B parameters)
- **Subset of the data** (3,227 train)
- **1 epoch**, **1 seed**, **1 prompt template**
- No hyperparameter search
- No downstream (scanner-level) metric
- Qualitative look at only $\mathbf{n = 3}$ samples

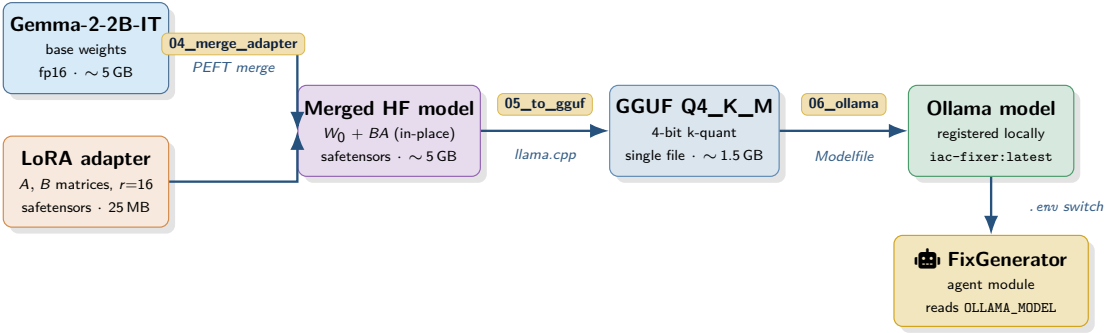
Threats to validity

- **External**: dataset skewed toward public Docker/Kubernetes repos.
- **Construct**: "smell" = "pattern detected by Checkov/tfsec/KICS"; excludes business-logic flaws.
- **Internal**: single run, no variance estimate.

What this run *does* demonstrate

The end-to-end pipeline runs within Kaggle's free GPU tier, produces a coherent 25 MB adapter, and surfaces a *measurable* obstacle to improvement: label noise.

Integration path: adapter → Ollama



Three new scripts added to the repo

- `04_merge_adapter.py` — `PeftModel.merge_and_unload()`
- `05_convert_to_gguf.sh` — llama.cpp `convert_hf_to_gguf.py` + `llama-quantize`
- `06_ollama_create.sh` — Modelfile + `ollama create iac-fixer`

Framework change required: one line

Set `OLLAMA_MODEL=iac-fixer` in `.env`. `FixGenerator` already reads the backend from env vars, so no code change is needed downstream.

Config B evaluation — the remaining deliverable

Evaluation plan (from Rapport 2, Chapter 6)

Config	Composition	Status
A	Checkov alone	**Measured**. F1=0.132, P=0.084, R=0.304
B	Analyzer + fine-tuned LLM + validator (full pipeline)	**Pending** — this work enables it
C	Config B + self-consistency (N=5) + iterative retrieval	Deferred

Primary metrics (consistent with Rapport 2)

- Patch Validity Rate
- Smell Elimination Rate
- No-New-Issues Rate
- Macro-F1 over the ground-truth smell set

Sequencing

1. Merge adapter locally
2. GGUF + Ollama registration
3. Run `scripts/evaluate.py` on the labelled 16-file benchmark
4. Compare Config A vs B

Roadmap beyond Config B

Data

- Hunk-level diff filter
- Scanner-fixture-only corpus
- Extend scraper v2 with hunk-level classifier

Model

- Try Qwen2.5-Coder-3B (non-gated, stronger on code)
- 2–3 epochs with early stopping
- Small HP grid search over r and lr
- Prompt-template diversity

Loop

- Close the RAG $\leftrightarrow$ validator loop
- Self-training on validator-failed examples
- Multi-seed replication for variance

The present run is one building block. Closing the loop requires the framework-level evaluation, which is the next concrete step.

Take-aways

What we can say

- A complete QLoRA training pipeline for IaC fix generation runs end-to-end on Kaggle's free T4 in $< 2\,\mathrm{h}$.
- A 25 MB adapter is produced from 3,227 scanner-validated training records; losses are stable, no overfitting signal.
- On three inspected samples, outputs are well-formed and contain idiomatic hardening patterns.

What we cannot say yet

- Anything about downstream quality on real scanner-level metrics — the Config B evaluation has not been run.
- Anything statistically meaningful — one seed, one config, three qualitative samples.

The binding constraint

Label noise in commit-scraped pairs. Fixing it (hunk-level filter, scanner fixtures) is the single most promising next step — more so than scaling the model.

Questions ?

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Based on Rapport 5 — `Rapports/drafts/rapport5/rapport5.pdf`